



Sustainability of the Traditional Adan Rice Farming System in the Border Region of North Kalimantan: Linking Land Characteristics to Local Food Security

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Abstract: Adan rice farming is a traditional agricultural system uniquely practised in the border region of North Kalimantan, Indonesia. Cultivated organically using buffalo manure and crop residues, Adan rice contributes to local food sovereignty and shows cross-border market potential in Malaysia and Brunei Darussalam. However, its productivity remains low, and its sustainability status has not yet been clearly established. This study aimed to determine the land characteristics of Adan rice paddies, assess the sustainability status of Adan rice farming, and identify sensitive attributes influencing its sustainability. Data collection involved laboratory analysis of soil samples and a sustainability assessment using the Multidimensional Scaling (MDS) approach with Rapid Appraisal of the Status of Farming (Rap-FARM) software, supported by leverage (sensitivity) analysis, Monte Carlo simulation, stress-value testing, and the coefficient of determination (R^2). The respondents consisted of 73 farmers in Krayan Sub-district. The results indicated that Adan rice paddies were dominated by clay loam, sandy clay loam, and sandy loam, with soil pH ranging from acidic to slightly acidic. Fertility indicators showed moderate to high organic carbon content, low to moderate nitrogen levels, low to moderate phosphorus levels, moderate potassium levels, and moderate to high cation exchange capacity (CEC). The novelty of this study lies in linking detailed land characteristics with a multidimensional sustainability framework for a rice cultivation system located in a strategically important border region. Unlike previous studies that focused primarily on high-yield commercial rice, this research highlights how local wisdom preserved in traditional rice cultivation can contribute to sustainable agricultural systems and food security in border areas.

Keywords: Adan rice; Indigenous farming systems; Sustainability assessment; Land characteristics; Border region; North Kalimantan

1. Introduction

Sustainable agriculture has been widely discussed, particularly about how agricultural practices can be defined and qualified as sustainable. Nevertheless, there is a consensus among scholars that agricultural sustainability is determined by three main dimensions: the economic dimension (profitable and viable production), the environmental dimension (ecological balance and conservation of natural resources), and the social dimension (resilient rural livelihoods). All of which are integral to the framework of sustainable development (Abubakar & Attanda, 2013; National Research Council, 2010; Pretty, 2008). These principles have since been extended to incorporate technological and institutional dimensions as integral components of sustainable agricultural development (Suyitman et al., 2009). Gomiero et al. (2011) emphasized that sustainable agriculture was essential for maintaining long-term agricultural productivity while simultaneously conserving natural resources, particularly soil fertility, water quality, and biodiversity. Similarly, Powlson et al. (2011) stated that sustainable agriculture enabled farming systems to continuously produce food without causing irreversible damage to ecosystems. In addition, Martin et al. (2019) highlighted that agricultural sustainability was closely linked to long-term food

security.

In Indonesia, where rice serves as the primary staple food, concerns regarding rice production have positioned sustainable agriculture as a central focus of agricultural policy (Salassa et al., 2025). New approach to achieving sustainable agriculture is realized through organic farming systems, which have the potential to increase farmers' income due to higher market prices and growing global consumer demand (Permatasari et al., 2021). However, the sustainability of organic agriculture remains limited due to high certification costs, complex cultivation standards, and certification processes that are often misaligned with local farming practices, thereby necessitating stronger governmental support policies. In addition to organic farming, sustainable rice cultivation is found in traditional farming systems that continue to apply local wisdom. Such practices include the use of local rice varieties, traditional irrigation systems, organic fertilizers, nature-based technologies, communal cooperation, and cultural beliefs that emphasize harmony between humans and nature. These practices contribute to ecological sustainability, local food security, and social resilience. Nevertheless, significant challenges persist, particularly in terms of marketing and pricing mechanisms (economic dimension) as well as relatively low production levels (technological dimension) (Dewi & Paramarta, 2025; Nurzakiah et al., 2024; Pratama et al., 2024; Rahman, 2024).

North Kalimantan Province, located in the northern part of Borneo and directly bordering Sarawak and Sabah (Malaysia), presents a strategic case for studying agricultural sustainability (Satyawan, 2018). The Krayan Sub-district of Nunukan Regency, which directly shares land borders with Sabah, is characterized by rich natural resources that support crop and livestock production. The majority of its population are farmers cultivating Adan rice, the region's main commodity, which has received Geographical Indication (GI) certification (ID G 000000013, January 6, 2012). Adan rice is highly valued for its distinctive taste, fine grain texture, and reliance on organic inputs such as buffalo manure and harvest residues (Mubarak et al., 2021).

As an indigenous variety cultivated for generations by the Dayak community, Adan rice has undergone natural adaptation, farmer-led selection, and genetic diversification. Currently, 15 to 20 landraces are recognized and they include Adan white (short and milky grains), Adan small (slender grains), Adan red (red-colored grains), and Adan black (dark grains) (Wahyuni et al., 2023). Despite its cultural and economic importance, the productivity of Adan rice averages only 3.5 t/ha, with relatively long cropping cycles of 6–7 months. Its price, around USD 12.33 per 15 kg, reflects its high market value, particularly in cross-border trade with Malaysia and Brunei Darussalam (Prasetyono et al., 2022).

However, the sustainability of Adan rice farming faces critical challenges, including low productivity, long harvest duration, limited farming infrastructure, lack of access to capital, and dependence on traditional cultivation techniques. Moreover, market access remains constrained as the Krayan Sub-district is geographically isolated and can only be reached by small aircraft from surrounding regions. These challenges threaten not only the economic viability of farmers but also the long-term ecological and social-cultural sustainability of the system.

Therefore, this study aims to determine the land characteristics of Adan rice fields and to assess the sustainability status of Adan rice farming from a multidimensional perspective, encompassing ecological, economic, social-cultural, and technological dimensions. By integrating soil physical and chemical properties with a multidimensional sustainability assessment, this study seeks to identify strategic interventions that could support sustainability and enhance the production of Adan rice. Furthermore, this research contributes to a scientific understanding of local rice farming systems in border regions and provides insights for developing sustainable agricultural strategies that could strengthen local food security and cross-border trade opportunities.

2. Methodology

2.1 Location and Time

The research was conducted in Krayan Sub-district, Nunukan Regency, North Kalimantan Province for 6 months starting from April to September 2024. The Krayan Sub-district has an area of 777.91 km² and is directly adjacent to the Western and Northern parts of Sarawak Malaysia (Figure 1). The indigenous and the majority population are the Dayak Lundayeh tribe. The population in 2024 was 3,456 people and the density was 4 people/km² (Nugraha et al., 2018). This study was divided into two components: Land Physical Assessment and Farm Sustainability Assessment.

2.2 Land Physical Assessment

The land physical assessment aimed to obtain data on the characteristics of Adan rice paddy fields through soil sampling. The materials and equipment used for soil sampling followed the basic instructions described by Wahyunto et al. (2016), including a Global Positioning System (GPS) device, administrative maps, soil maps, soil pH tester, soil auger, clinometer, plastic buckets, permanent markers, plastic bags, machete, and laboratory-grade chemicals for soil sample analysis to identify soil physio-chemical parameters. Soil samples were collected randomly from eight points within the rice fields. Samples were taken from the plow layer (root zone) at a depth

of 10–30 cm using a soil auger. From each sampling point, 250 g of soil was collected and labeled in plastic bags. The collected soil was then composited, clear of root residues, and homogenized in a container to obtain a representative soil sample. The physical and chemical properties of the soil, covering soil texture loam, sandy loam, and sandy clay loam, pH, organic carbon (organic C), total nitrogen (N), available phosphorus (P_2O_5 , extracted with 25% HCl), available potassium (K_2O , extracted with 25% HCl) and cation exchange capacity (CEC) were analyzed at the Soil Laboratory, Faculty of Agriculture, Universitas Borneo Tarakan. The analyzed physical property was limited to soil texture while the chemical properties were assessed based on the Soil Chemical Property Evaluation Criteria (Hardjowigeno, 2003), as shown in Table 1.

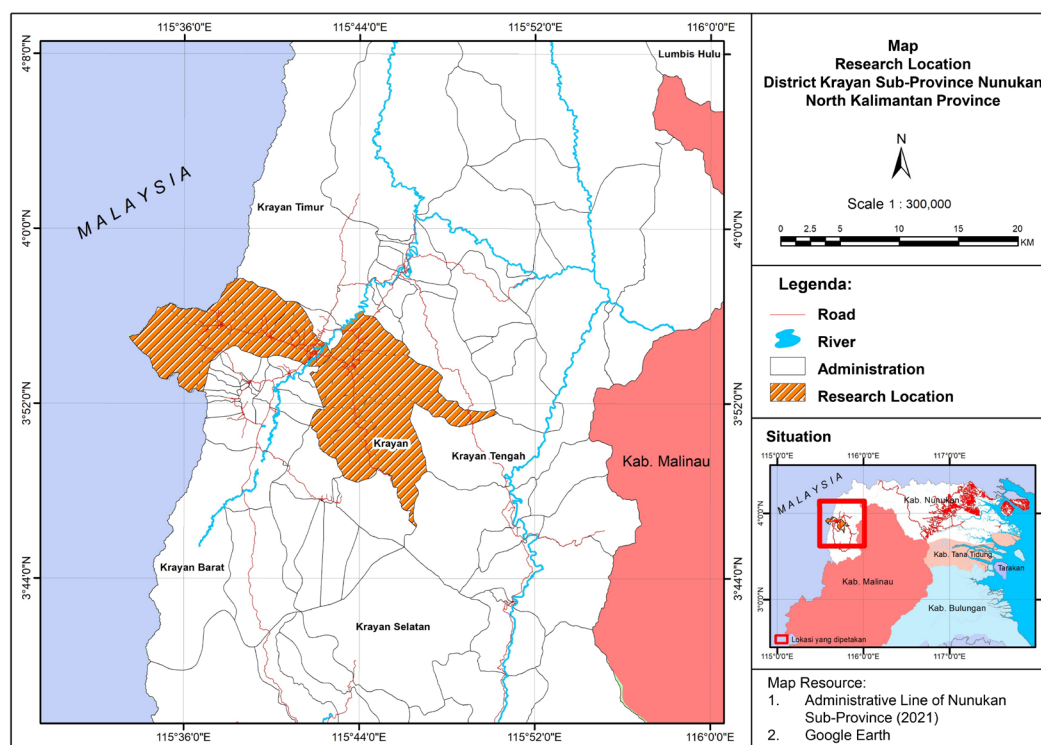


Figure 1. Study area in Krayan Sub-district, Nunukan Regency, North Kalimantan Province, Indonesia

Table 1. Criteria for evaluating the properties of soil chemicals

Property	Very Low	Low	Moderate	High	Very High	
Organic C (%)	<1.00	1.00–2.00	2.01–3.00	3.01–5.00	>5.00	
Total N (%)	<0.10	0.10–0.20	0.21–0.40	>0.40	—	
Avail. P ₂ O ₅ (25% HCl) (mg/100 g)	<10	10–20	21–40	41–60	>60	
Avail. K ₂ O (25% HCl) (mg/100 g)	<10	10–20	21–40	41–60	>60	
CEC (cmol/kg)	—	<10	10–20	>20	—	
Property	Very Acidic	Acidic	Slightly Acidic	Neutral	Slightly Alkaline	Alkaline
pH (H ₂ O)	<4.5	4.5–5.5	5.6–6.5	6.6–7.5	7.6–8.5	>8.5

Note: Organic C = organic carbon; Total N = total nitrogen; Avail. P_2O_5 = available phosphorus extracted with 25% HCl; Avail. K_2O = available potassium extracted with 25% HCl; CEC = cation exchange capacity; classification criteria adapted from Hardjowigeno (2003).

2.3 Farm Sustainability Assessment

This part of the study aimed to evaluate the sustainability status and identify sensitive attributes in the Adan rice farming system. The selection of expert respondents was based on the purposive sampling method, which is a method of determining respondents with criteria set by the researcher (Oribhabor & Anyanwu, 2019). A total of four expert respondents were obtained under the concerned study and they consisted of two extension expert workers of agriculture, one sub-district staff, and one tribal chief. With simple random sampling method, the farmer respondents were selected from those who cultivated Adan Rice. Based on the preliminary data from agricultural extension workers, there were 240 farmers; among them, 73 (30%) were chosen as stated by Arikunto

(2010). If the population members were more than 100, 10–20% could be taken as a sample.

Sustainable analysis was conducted by using the Multidimensional Scaling (MDS) approach which included ecological, economic, social-cultural, and technological dimensions. The MDS method is a statistical technique transforming multidimensional into simpler dimensions with the technique of Rapid Appraisal for Fisheries (Rapfish) developed by the Fisheries Center, University of British Columbia (Pitcher et al., 2013; Pratama et al., 2024; Wahyuni & Santoso, 2023). In this study, the approach with Rapid Appraisal of the Status of Farming (Rap-FARM) was adopted from Rapfish.

The determination of index and sustainable status, according to Below et al. (2012), started by determining the attributes and scores for each dimension. Factors for consideration included authentic data from the field, observations and initial interviews, references from the literature and judgment from researchers and their scientific assumptions. Categories of sustainability status can be viewed in Table 1. The score values of the attributes of each sustainable dimension were analyzed in the Rapfish program in Microsoft Excel to obtain a sustainable index value. The sustainability index was interpreted using the categories shown in Table 2: analysis with the Rapfish program on Microsoft Excel would simultaneously perform sensitive analysis (leverage analysis), Monte Carlo, stress value, and coefficient of determination (R^2). Sensitive analysis (leverage analysis) was used to obtain sensitive attributes that could affect the status of sustainability and observe the priority order based on the changes in the value of root mean square (RMS). The greater the RMS value, the more sensitive/larger role of these attributes is to improve the sustainable status of Adan rice farming and the intervention or improvement that needs to be done.

Table 2. Range of values of sustainability status

No.	Index	Category
1	0–25.00	Not sustainable
2	25.01–50.00	Less sustainable
3	50.01–75.00	Moderately sustainable
4	75.01–100	Sustainable

Source: Kavanagh & Pitcher (2004)

Monte Carlo analysis was conducted to estimate the level of random error in the model generated through the MDS analysis of all dimensions at the 95% confidence level. The results of sustainability index analysis of the MDS and Monte Carlo were compared. According to Girard & Cliff (1976), if the difference between the two values was small or less than 25%, it could be concluded that (1) The analysis process carried out repeatedly was stable; (2) Data entry errors and missing data could be avoided; (3) Errors in scoring each attribute were relatively small; and (4) The modified Rapfish method used was appropriate for evaluating sustainability in the conducted research or the results were close to the actual situation and the data analyzed had been mapped properly. Furthermore, the stress value represents a measure of lack of fit, where higher stress values indicate a poorer fit of the selected attributes in the MDS analysis. The stress value and R^2 are used to assess the adequacy of the constructed model based on the selected attributes. These indicators determine whether additional attributes are required and whether the selected dimensions appropriately represent the actual conditions in the field. The model or analytical results could be considered acceptable when the stress value is less than 0.25 and the R^2 value approaches 1.

3. Results and Discussions

3.1 Characteristics of Adan Rice Paddy Fields

The characteristics of Adan rice paddy fields are closely related to the availability of nutrients for plant growth, as reflected in the physical and chemical properties of the soil cultivated by farmers, as presented in Table 3.

The soils in the Adan rice paddy fields were classified as clay loam, sandy clay loam, and sandy loam, which are favorable for rice cultivation due to their high water-holding capacity and good aeration. Such textures ensure adequate oxygen availability and nutrient retention, supported by relatively high CEC (Hardjowigeno, 2003). The CEC values at the study site ranged from moderate to high, as they were largely influenced by the dominance of clay fractions that provided a large colloidal surface area (Arifien & Anggarawati, 2019).

Soil organic C ranged from 2.71% to 4.25%, corresponding to moderate to high levels across the study site, indicating substantial organic matter contributions from plant and animal residues. The decomposition of organic matter improves soil structure, enhances the availability of nutrients such as total nitrogen (total N), available phosphorus (Avail. P_2O_5), available potassium (Avail. K_2O), and other micronutrients, and increases both water retention and CEC (Munawar, 2011; Zulfa & Bowo, 2023).

Soil pH (H_2O) ranged from acidic to slightly acidic (4.92–5.90), a condition that might reduce macronutrient availability and increase the solubility of toxic elements such as iron (Fe) and aluminum (Al). To improve soil

fertility, liming with dolomite and the incorporation of compost were recommended as they enhanced the decomposition of organic matter, reduced exchangeable Al, and gradually increased soil pH (Dawid, 2021).

The availability of nutrients varied across the site: Avail. P₂O₅ ranged from 17.66 to 36.14 mg/100 g, corresponding to low to moderate levels, while Avail. K₂O ranged from 21.16 to 36.42 mg/100 g and was consistently classified as moderate. These conditions were likely influenced by parent material deficiency in base cations and high rainfall, which promoted nutrient leaching (Arifien & Anggarawati, 2019). Total N ranged from 0.17% to 0.27%, corresponding to low to moderate levels. Nitrogen remained the most limiting nutrient due to its susceptibility to leaching, volatilization, and rapid plant uptake, thus emphasizing the importance of incorporating residues to sustain soil fertility (Patti et al., 2018; Rahmadani et al., 2020; Sari et al., 2022).

Table 3. Physical and chemical characteristics of Adan rice paddy fields

Land Unit	pH (H ₂ O)	Organic C (%)	Total N (%)	Avail. P ₂ O ₅ (25% HCl) (mg/100 g)	Avail. K ₂ O (25% HCl) (mg/100 g)	CEC (cmol/kg)	Texture Class
KSS-1	5.45 Acidic	3.02 High	0.23 Moderate	33.24 Moderate	24.63 Moderate	25.38 High	Clay loam
KSS-2	5.72 Slightly acidic	3.65 High	0.19 Low	28.29 Moderate	35.08 Moderate	24.21 High	Sandy clay loam
KSS-3	5.33 Acidic	2.71 Moderate	0.27 Moderate	30.44 Moderate	33.19 Moderate	18.28 Moderate	Sandy loam
KSS-4	5.16 Acidic	4.01 High	0.22 Moderate	17.66 Low	27.84 Moderate	19.30 Moderate	Sandy loam
KSS-5	4.92 Acidic	4.25 High	0.20 Low	21.30 Moderate	36.42 Moderate	19.53 Moderate	Sandy clay loam
KSS-6	5.79 Slightly acidic	3.55 High	0.17 Low	22.14 Moderate	36.10 Moderate	20.88 High	Sandy clay loam
KSS-7	5.66 Slightly acidic	3.92 High	0.17 Low	20.74 Low	21.16 Moderate	17.90 Moderate	Sandy clay loam
KSS-8	5.90 Slightly acidic	3.65 High	0.22 Moderate	36.14 Moderate	28.53 Moderate	20.29 High	Sandy clay loam

Note: Organic C = organic carbon; Total N = total nitrogen; Avail. P₂O₅ = available phosphorus extracted with 25% HCl; Avail. K₂O = available potassium extracted with 25% HCl; CEC = cation exchange capacity; KSS = Krayan soil sampling point.

3.2 Index Values and Sustainability Status

The sustainability status of Adan rice farming based on ecological, economic, social-cultural, and technological dimensions could be explained through the index values in Table 4 and illustrated in Figure 2.

Table 4. Sustainability index values and status for each analyzed dimension of Adan farming

Dimensions	Sustainability Index	Sustainability Status
Ecological	63.01	Moderately sustainable
Economic	48.95	Less sustainable
Social-cultural	60.19	Moderately sustainable
Technological	25.77	Less sustainable
Multidimensional	49.48	Less sustainable

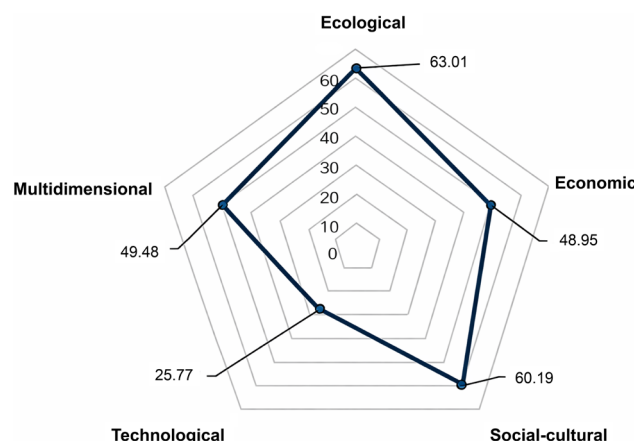


Figure 2. Radar chart of sustainability index values for each dimension

The multidimensional index value for the sustainability of Adan rice farming was 49.48, as shown in Figure 2, which falls into the less sustainable category. This result was obtained from the assessment of attributes for four dimensions of sustainability: the ecological dimension had ten attributes; the economic dimension had five attributes; the social-cultural dimension had five attributes; and the technological dimension had five attributes. Two dimensions were categorized as moderately sustainable, namely the ecological and social-cultural dimensions, while the economic and technological dimensions were categorized as less sustainable. The varied index values and sustainability status across dimensions indicated that multidimensional sustainability could be improved if interventions for the development of Adan rice farming were carried out holistically across all dimensions.

Table 4 shows the sustainability index in Adan rice farming and provides important insights into the overall status of sustainability in various dimensions. These results reflected the complex interplay between ecological, economic, social-cultural, and technological aspects in this agricultural context. The ecological dimension, with an index value of 63.01, indicated that Adan rice farming had a relatively good performance in conserving local ecosystems and natural resources. This illustrated that farmers had implemented practices that supported environmental balance and resource maintenance, to reduce negative impacts on the local ecology. However, the economic dimension with an index value of 48.95 revealed potential improvements in the financial aspects of farming. This indicated that economic management, production efficiency, and income diversification still needed to be improved. The possibility of more stable earnings and better management of economic risk could be a focal point for improvement in this dimension. The social-cultural dimension, with an index value of 60.19, indicated that relations between farmers and other social aspects tended to be quite sustainable.

The practice of cooperation and community bonding may have supported the involvement of farmers in this farming. However, further involvement in community development and empowerment could help strengthen this social-cultural dimension. The most striking result was the technological dimension with a low index value, namely 25.77. This showed that the potential for using more advanced technology in Adan rice farming had not been maximized. Implementing modern technologies, such as smart irrigation or more efficient crop management systems, could increase productivity and reduce environmental impact. The overall multidimensional index value was 49.48, which indicated a less sustainable status. Variability in index scores across dimensions emphasizes the need for an integrated approach to improvement. The steps forward should embrace all dimensions of sustainability to achieve better overall results. In order to achieve better sustainability, there have to be collaborative efforts among farmers, government, and other related parties (Zhang et al., 2016). Coordinated interventions in economic, technological, and social aspects could help achieve a better balance in Adan rice farming. By leveraging the strengths of each dimension, sustainability goals could be realized through holistic actions that prioritize ecological, economic, social-cultural, and technological balance (Chan et al., 2020).

As shown in Table 5, the stress values for all five dimensions were below 0.25, while the R^2 values were close to 1. These results indicate that the dimensions and attributes used are sufficient and the accuracy of the dimensions being studied could reflect the actual situation in the field. According to Kavanagh & Pitcher (2004), the analysis results could be rated quite good if the stress value was less than 0.25 (25%) and the R^2 was close to 1.

Table 5. Statistical parameters of the Multidimensional Scaling (MDS) analysis for the sustainability of Adan rice farming

Statistical Parameters	Dimensions				
	Ecological	Economic	Social-Cultural	Technological	Multidimensional
Stress	0.13	0.15	0.14	0.15	0.14
R^2	0.94	0.94	0.95	0.94	0.94

As presented in Table 6, the differences between the MDS and Monte Carlo sustainability index values at the 95% confidence level were all less than 1, indicating a high level of confidence in the analysis results (Briggs et al., 1999). The small differences also suggest that (1) the scoring of each attribute had a relatively small error; (2) variation in scoring due to differences in opinion was limited; (3) the process of the MDS analysis carried out repeatedly was relatively stable; and (4) data entry errors and missing data were minimal.

Table 6. Difference between the Multidimensional Scaling (MDS) and Monte Carlo sustainability index values at the 95% confidence level

Method	Dimensions				
	Ecological	Economic	Social-Cultural	Technological	Multidimensional
MDS	63.01	48.95	60.19	25.77	49.48
Monte Carlo	62.51	48.81	59.31	25.93	49.14
Difference	0.50	0.14	0.88	0.16	0.34

3.3 Sensitive Attributes

3.3.1 Ecological dimension

As shown in Figure 3, the leverage analysis of the ecological dimension identified several sensitive attributes affecting the sustainability of Adan rice farming, including planting of other crops, pest and disease attacks, two-year crop yield, straw waste utilization, the use of chemical pesticides, the use of chemical fertilizers, seed origin, water source, land fertility, and distance from land to settlements. Among these, the use of chemical pesticides showed the highest RMS value, indicating that it was the most influential leverage factor in the ecological dimension. Other important attributes included the use of chemical fertilizers, seed origin, two-year crop yield, and straw waste utilization, while lower-ranked attributes had a relatively smaller influence on the ecological sustainability index.

The attribute of chemical pesticide use as a leverage factor reflects the ecological importance of farming practices in Adan rice cultivation. In practice, Adan rice farmers tend to use little or no chemical pesticides, which represents an important ecological advantage (Wahyuni & Santoso, 2023). This condition is further supported by several other attributes that may strengthen ecological sustainability, including limited dependence on chemical fertilizers, the use of self-produced seeds, biological pest and disease management, and the utilization of straw waste to maintain soil fertility. In addition, buffalo manure and urine left in the fields during the fallow period may also contribute to nutrient cycling and reduce dependence on external chemical inputs. These practices are consistent with the principles of organic farming, which emphasize the avoidance of synthetic chemical pesticides and fertilizers, the use of natural materials, and the maintenance of soil fertility and productivity through biological and ecological approaches (Gamage et al., 2023; Yuriansyah et al., 2020). Therefore, encouraging Adan rice farming to obtain organic labeling could serve as an important intervention strategy for local government, as it would not only support ecological sustainability but also increase the market value and competitiveness of Adan rice.

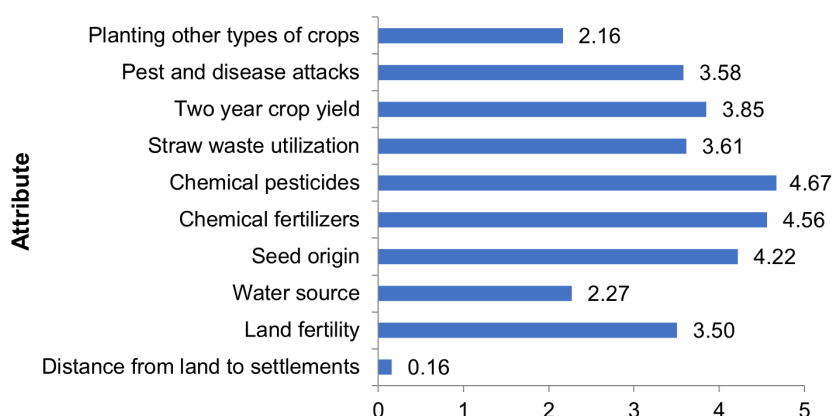


Figure 3. Leverage analysis (root mean square [RMS] change) of ecological attributes in Adan rice farming

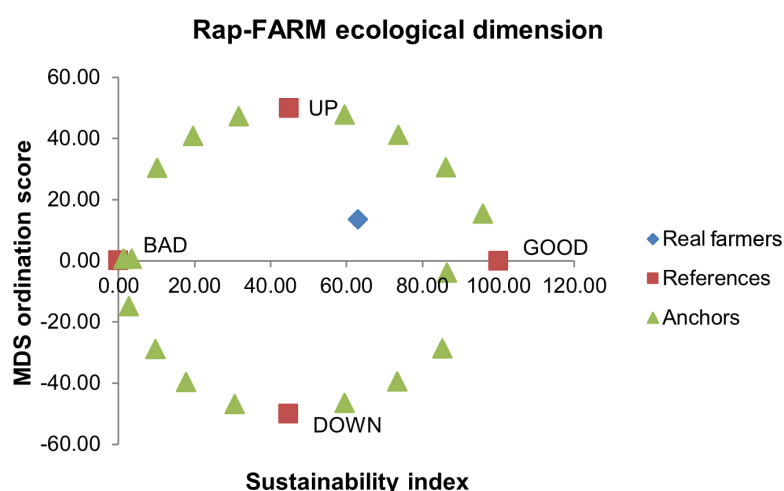


Figure 4. Monte Carlo analysis of the Multidimensional Scaling (MDS) ordination for the ecological dimension of Adan rice farming

The stability of the ecological sustainability ordination is further illustrated in Figure 4, which presents the Monte Carlo analysis of the MDS results. The close clustering of the ordination points indicates that the ecological sustainability assessment was relatively stable and not highly sensitive to random error or scoring variation. This result is consistent with the small difference between the MDS and Monte Carlo sustainability index values, suggesting that the ecological dimension was represented reliably in the sustainability assessment. It also indicates that scoring errors, differences in judgment, and data entry bias were minimal. Therefore, the ecological sustainability results can be considered sufficiently robust for identifying priority intervention attributes in Adan rice farming.

3.3.2 Economic dimension

Figure 5 shows that the economic sustainability of Adan rice farming was influenced by five main attributes: (1) the price of rice at the farm level for the last 3 years, (2) the average income of farmers compared to the minimum wage in the Regency, (3) partnerships (regular buyers), (4) the market for selling crops, and (5) the status of farmland ownership. Among these, market access had the highest RMS value, indicating that it was the most sensitive attribute in the economic dimension. The price of rice reflected external factors influencing farmers' income while the average income of farmers compared to the minimum wage provided an overview of farmers' economic welfare. Partnerships with regular buyers could provide income stability while the market for selling harvests determined farmers' access to markets and profit potential. Land ownership status also affected income sources and control over agricultural output.

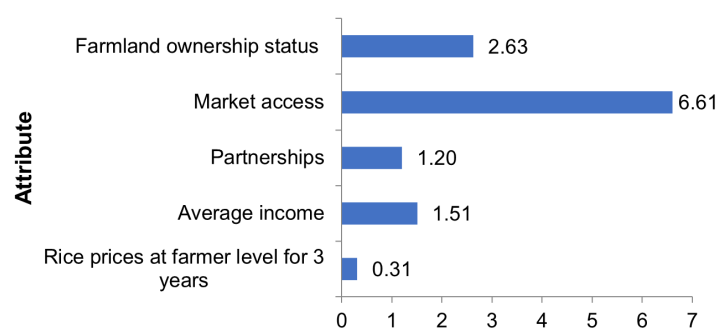


Figure 5. Leverage analysis (root mean square [RMS] change) of economic attributes in Adan rice farming

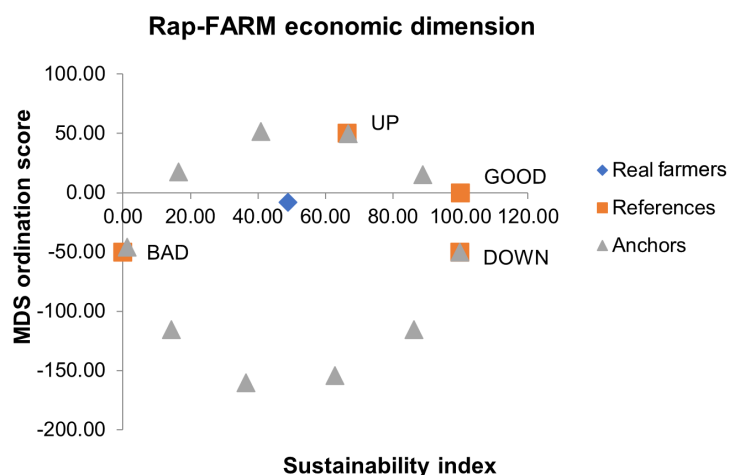


Figure 6. Monte Carlo analysis of the Multidimensional Scaling (MDS) ordination for the economic dimension of Adan rice farming

The analysis of the economic dimension showed that the main economic constraint in Adan rice farming lies not only in production, but also in commercialization. Farmers generally do not have a fixed market channel for their harvests. Instead, grain is often stored and sold only when buyers are available. Such uncertainty in marketing reduces income stability and may partly explain why farmers tend to cultivate rice only once a year. In this context, improving market access is likely to be a key intervention for strengthening the economic sustainability of Adan rice farming. As noted by MacDonald et al. (2007), the economic dimension emphasizes optimal profit without reducing economic potential for future generations. Therefore, local government support in promoting and marketing Adan rice, especially outside Krayan Sub-district and in neighboring countries, could help farmers

secure more stable incomes and better recover production costs and labor inputs.

Figure 6 shows that the Monte Carlo results for the economic dimension were stable and suitable for interpreting the economic sustainability of Adan rice farming.

3.3.3 Social-cultural dimension

As shown in Figure 7, nine attributes included to characterize the social-cultural dimension were (1) conflicts between farmers requiring land or water for farming, (2) the involvement of family members as labor, (3) livelihoods other than Adan rice farming, (4) the existence to preserve water and land resources, (5) the working together in farming, (6) the use of buffaloes to help fertilize land, (7) the number of buffaloes owned, (8) the feasibility of Adan rice farming, and (9) willingness to sell land. These attributes reflect important aspects of the social and cultural conditions surrounding Adan rice farming, including household livelihood strategies, social relations, traditional practices, and perceptions of agricultural viability and land use.

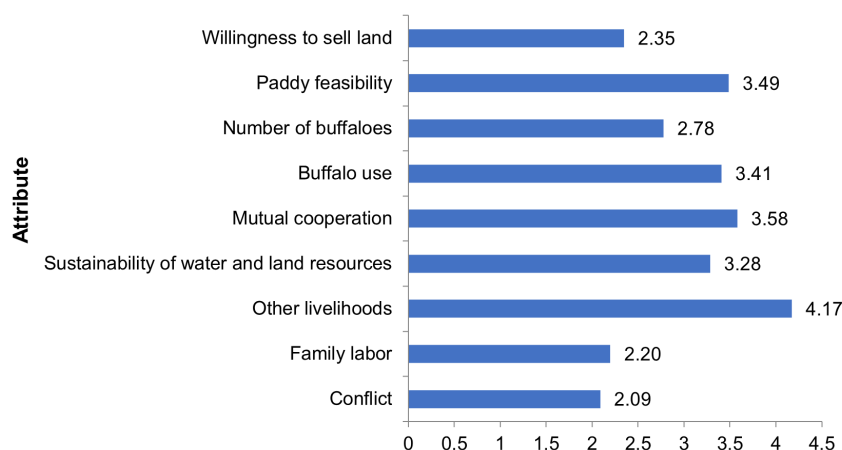


Figure 7. Leverage analysis (root mean square [RMS] change) of social-cultural attributes in Adan rice farming

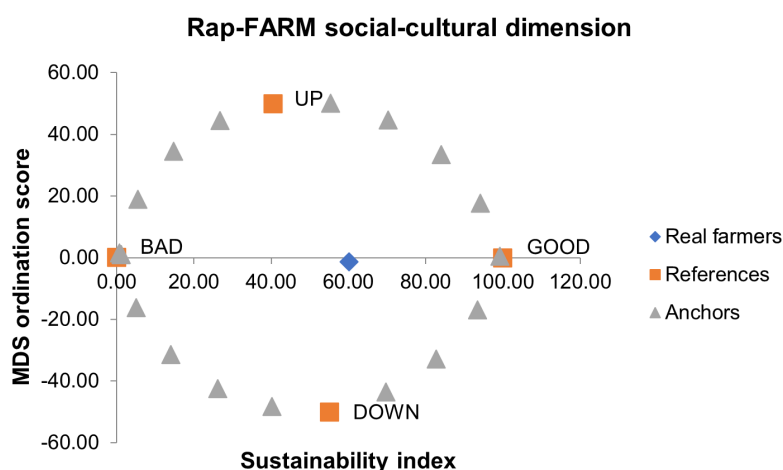


Figure 8. Monte Carlo analysis of the Multidimensional Scaling (MDS) ordination for the social-cultural dimension of Adan rice farming

The leverage analysis indicated that all nine attributes contributed to the sustainability of the social-cultural dimension, although with different levels of influence. Among them, other livelihoods had the highest RMS value (4.17), indicating that it was the most sensitive attribute in this dimension. This suggests that the availability or absence of income sources outside Adan rice farming plays a major role in shaping socio-cultural sustainability. The next most influential attributes were mutual cooperation (3.58), paddy feasibility (3.49), and buffalo use (3.41), all of which showed relatively high RMS values and therefore made substantial contributions to the sustainability index. The importance of other livelihoods reflects the dependence of many farmers on Adan rice farming as their primary or sole source of income. In practice, most farmers have limited alternative livelihood options and rarely cultivate other crops, making Adan rice farming central to household subsistence and local identity. This finding is consistent with previous studies showing that in traditional agricultural systems shaped by local wisdom,

farmland functions not only as a productive asset but also as an integral part of the social fabric of rural communities (Nugrahapsari et al., 2021; Nurzakayah et al., 2024; Rahmah, 2017).

Other sensitive attributes further highlight the social-cultural significance of Adan rice farming. Mutual cooperation (*gotong royong*) strengthens social cohesion and supports labor sharing within the farming community. Buffalo use and the number of buffaloes owned remain important, both as practical agricultural resources and as indicators of access to productive assets. In addition, the perceived feasibility of Adan rice farming influences farmers' willingness to continue cultivation, while the sustainability of water and land resources reflects local knowledge and long-term environmental stewardship (Barchia et al., 2021; Siwar et al., 2014). By contrast, attributes such as conflict, family labor, and willingness to sell land showed relatively smaller leverage values, although they still contributed to the overall social-cultural sustainability of the farming system. The potential for willingness to sell land, however, may threaten the continuity of traditional farming practices and local livelihoods.

Figure 8 indicates that the social-cultural sustainability estimates were also stable, with limited variation across repeated simulations.

3.3.4 Technological dimension

As revealed in Figure 9, the technological dimension of Adan rice farming was influenced by five main attributes: road condition, farming facilities, farming infrastructure, application of new technology, and the use of agricultural tools and machinery. Among these, farming infrastructure had the highest RMS value (11.94), indicating that it was the most sensitive attribute in the technological dimension. The second most influential attribute was road condition (8.56), followed by application of new technology (7.38), use of agricultural tools and machinery (6.55), and farming facilities (5.58). Although farming facilities had the lowest RMS value, they still contributed to the overall technological sustainability of Adan rice farming.

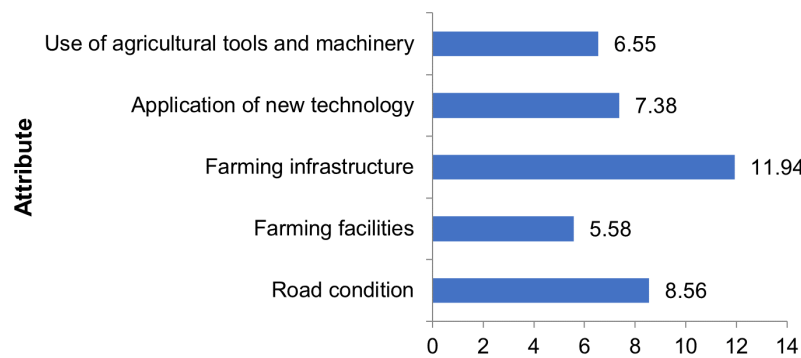


Figure 9. Leverage analysis (root mean square [RMS] change) of technological attributes in Adan rice farming

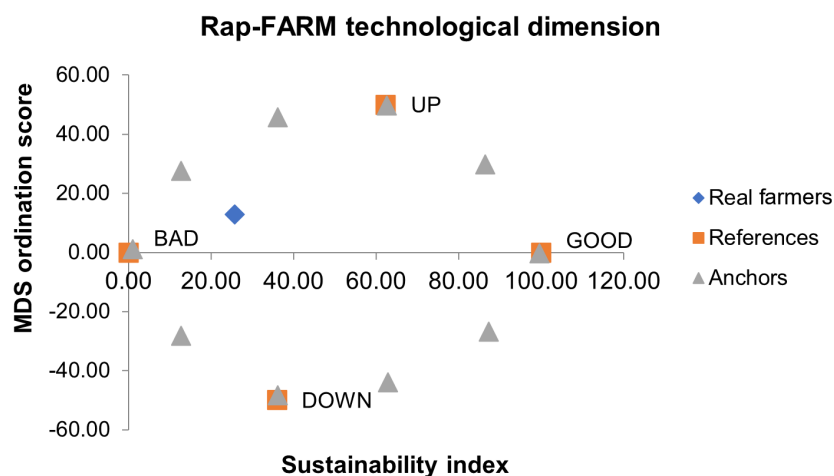


Figure 10. Monte Carlo analysis of the Multidimensional Scaling (MDS) ordination for the technological dimension of Adan rice farming

These results indicate that technological sustainability in Adan rice farming is determined primarily by

infrastructure and accessibility (e.g., irrigation systems, drainage, field access structures), followed by technology adoption and mechanization support. In practice, farming infrastructure remains limited, particularly because there are no permanent irrigation channels. Harvested grain is generally stored at home, and farmers rarely have access to dedicated storage facilities or nursery infrastructure. Although Krayan Sub-district has a main road in relatively good condition, farm access roads between settlements and rice fields are often muddy and slippery during the rainy season, reducing mobility and operational efficiency. These conditions show that technological sustainability is strongly influenced by structural infrastructure and accessibility factors.

The adoption of appropriate technologies and agricultural machinery can improve efficiency and support the sustainability of farming systems, but in Adan rice farming such adoption remains limited (Liang & Shah, 2023; Yunianti et al., 2022). In general, farming activities are still carried out traditionally, and farmers continue to rely heavily on manual tools for cultivation, harvesting, and post-harvest activities. Although equipment such as hand tractors and rice threshers has been provided by relevant government agencies, many farmers are reluctant to use them because they are considered impractical or not well suited to local conditions. This suggests that improving the application of new technology in Adan rice farming requires not only the provision of equipment, but also continuous education, technical assistance, and infrastructure support to increase adoption and efficiency. Accordingly, interventions should prioritize infrastructure, road access, farming facilities, and appropriate location-specific technologies, which are critical to enhancing the productivity and sustainability of Adan rice farming (Garb & Friedlander, 2014; Siwar et al., 2014).

Figure 10 further confirmed the stability of the technological dimension, suggesting that the MDS results were sufficiently robust for interpretation.

4. Conclusions

4.1 Summary of the Study

Adan rice farming represents a site-specific agricultural system practiced in the border region of North Kalimantan and can be considered a form of organic farming because it excludes the use of chemical fertilizers and pesticides. The land characteristics of Adan rice paddies indicate soil conditions that are generally suitable for rice cultivation, with soil textures classified as clay loam, sandy clay loam, and sandy loam. The soils showed moderate to high CEC, moderate to high organic C, and low to moderate levels of essential nutrients, including nitrogen, phosphorus, and potassium. However, the soil pH was acidic to slightly acidic, indicating the need for dolomite application to improve soil reaction. Overall, the sustainability status of Adan rice farming was classified as moderately sustainable in the ecological and social-cultural dimensions, but less sustainable in the economic and technological dimensions. The low sustainability performance in the economic dimension was mainly influenced by the absence of a stable and assured market for harvested rice. Meanwhile, the technological dimension was strongly affected by the limited availability of farming infrastructure, the condition of farm access roads, the low adoption of new technologies, and the limited use of agricultural tools and machinery.

4.2 Recommendations

Improving the sustainability of Adan rice farming requires multidimensional interventions covering ecological, economic, social-cultural, and technological aspects. These interventions should include maintaining existing organic farming practices, ensuring premium prices for Adan rice compared with other rice varieties, strengthening market access and marketing partnerships, and promoting the adoption of environmentally friendly technologies that are compatible with local culture and environmental conditions. In addition, support for infrastructure, road access, storage facilities, and location-specific technologies is necessary to improve productivity and long-term sustainability. Such efforts are expected to strengthen the contribution of Adan rice farming to food security, particularly in North Kalimantan.

Author Contributions

E.W. and D.E.C.C.W. conceived and designed the analysis. D.S. and S.E. collected the data. E.W. and D.E.C.C.W. contributed data or tools for analysis. D.S. and S.E. performed the analysis. D.S. interpreted the results. E.W., D.E.C.C.W., D.S., and S.E. wrote the paper. All authors have read and agreed to the published version of the manuscript.

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Informed Consent Statement

Informed consent was obtained from all subjects involved in the study.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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